Numerical modeling of laminar flow over a porous cylinder under endothermic steam methane reforming reaction

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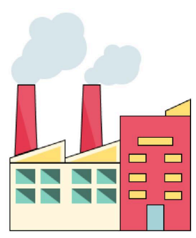
Motivation

Steam methane reforming (SMR) industrial relevance



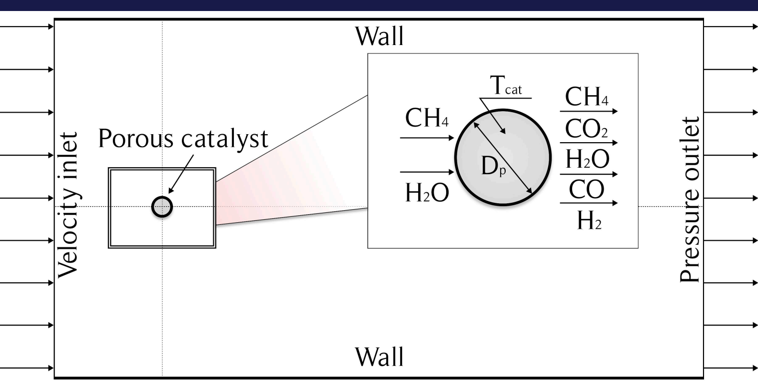
Lack of data on the effect of SMR on vortex dynamics

Computational Fluid Dynamics (CFD) Simulation

CFD

Motivation:
1. Steam methane reforming interaction with fluid dynamics in transient flows remains unexplored.
2. There is lack insights into how endothermic SMR reactions alter vortex shedding and heat transfer under non-isothermal conditions.
3. Understand the effect of chemical reactions on Karmán vortex street and heat transfer coefficient

Methodology



Key points:
1. A two-dimensional Reynolds-Averaged Navier-Stokes (RANS) approach was employed to simulate laminar flow and vortex dynamics, validated against literature data despite the inherent three-dimensional nature of real-world vortex shedding.
2. Chemical Kinetics Integration: User-Defined Functions (UDFs) in ANSYS Fluent were implemented to model steam methane reforming kinetics using Hou's reaction rate equations , adsorption coefficients, and equilibrium constants
3. Porous Media Modeling: Catalyst porosity (ε = 0.44) and permeability ($\alpha = 2.8\times10^{-14}$ m²) were incorporated via a momentum source term in the Navier-Stokes equations, simplifying fluid-catalyst interactions.

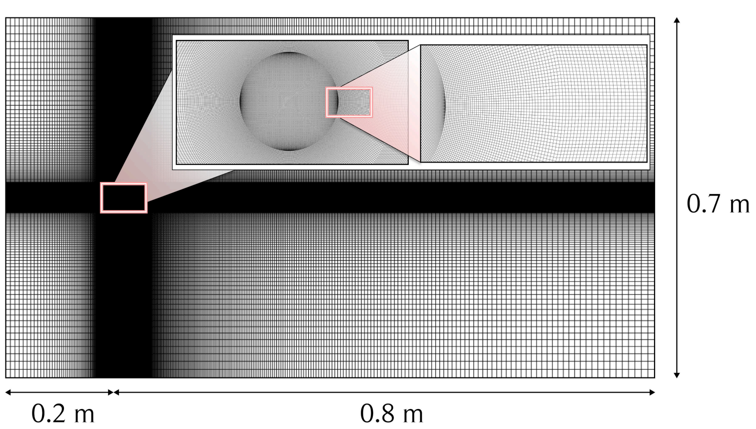
$$R_1 = \frac{k_1(p_{CH_4}p_{H_2O}^{0.5} - p_{H_2}^3 p_{CO}/K_1 p_{H_2O}^{0.5})}{p_{H_2}^{1.25}\cdot \mathrm{DEN}^2}$$

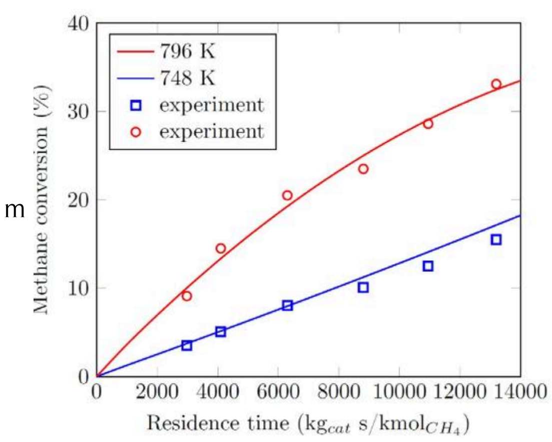
$$R_2 = \frac{k_2(p_{CO}p_{H_2O}^{0.5} - p_{H_2}p_{CO_2}/K_2 p_{H_2O}^{0.5})}{p_{H2}^{0.5}\cdot \mathrm{DEN}^2}$$

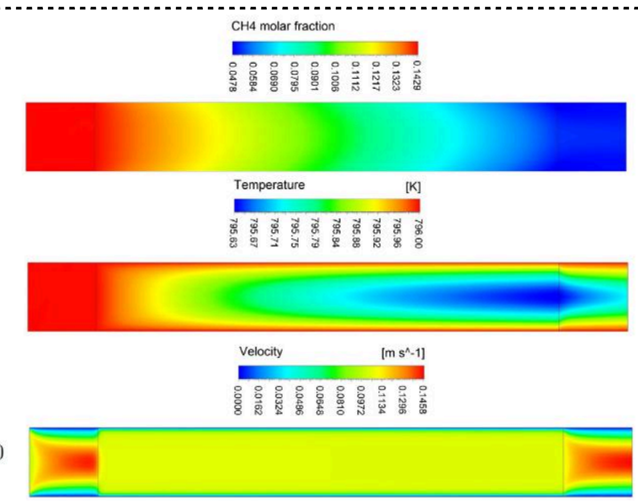
$$R_3 = \frac{k_3(p_{CH_4}p_{H_2O} - p_{H_2}{}^4 p_{CO_2}/K_3 p_{H_2O})}{p_{H_2}^{3.5}\cdot \mathrm{DEN}^2}$$

$$\mathrm{DEN} = 1 + K_{CO}\cdot p_{CO} + K_H\cdot p_{H_2}^{0.5} + K_{H_2O}\cdot p_{H_2O}/p_{H_2}$$

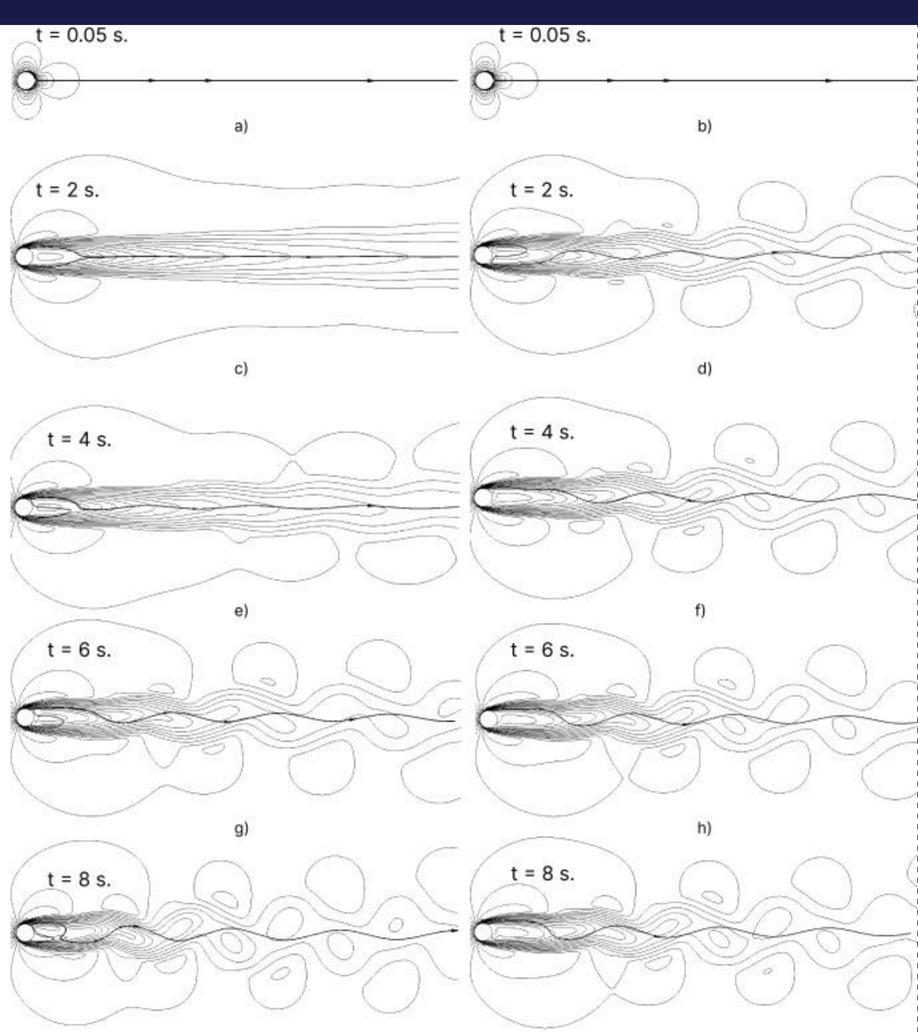
Kinetics		
k_1	=	$5.922\cdot 10^8 \exp\left(\frac{-209200}{R_g T}\right)$
k_2	=	$6.028\cdot 10^{-4} \exp\left(\frac{-15400}{R_g T}\right)$
k_3	=	$1.093\cdot 10^3 \exp\left(\frac{-109400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251\cdot \exp\left(\frac{-15900}{R_g T}\right)$
K_{H_2}	=	$5.68\cdot 10^{-10} \exp\left(\frac{93400}{R_g T}\right)$
K_{CO}	=	$5.127\times 10^{-13} \exp\left(\frac{140000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198\cdot 10^{17} \exp\left(\frac{-26830}{T}\right)$
K_2	=	$1.767\cdot 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117\cdot 10^{15} \exp\left(\frac{-22430}{T}\right)$

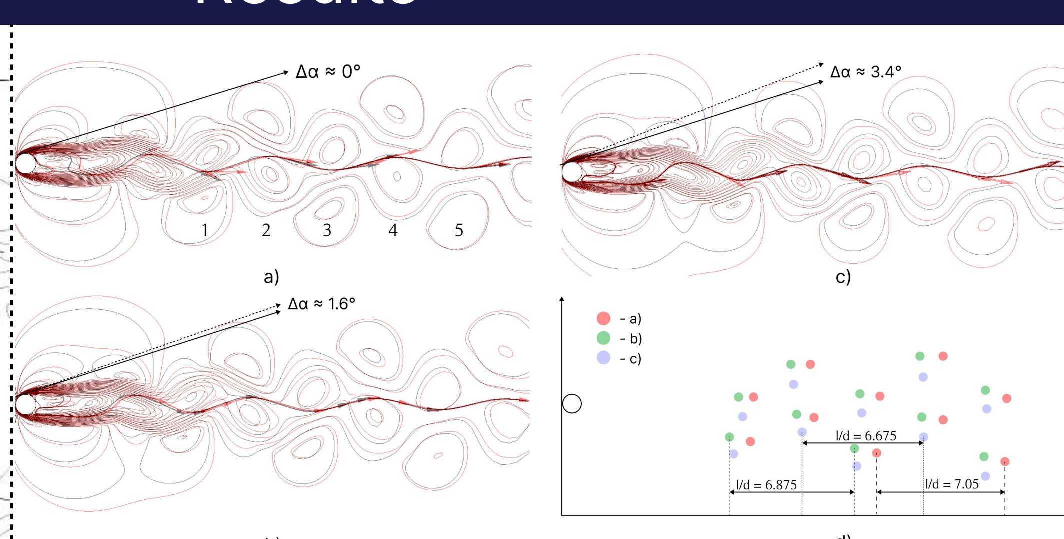


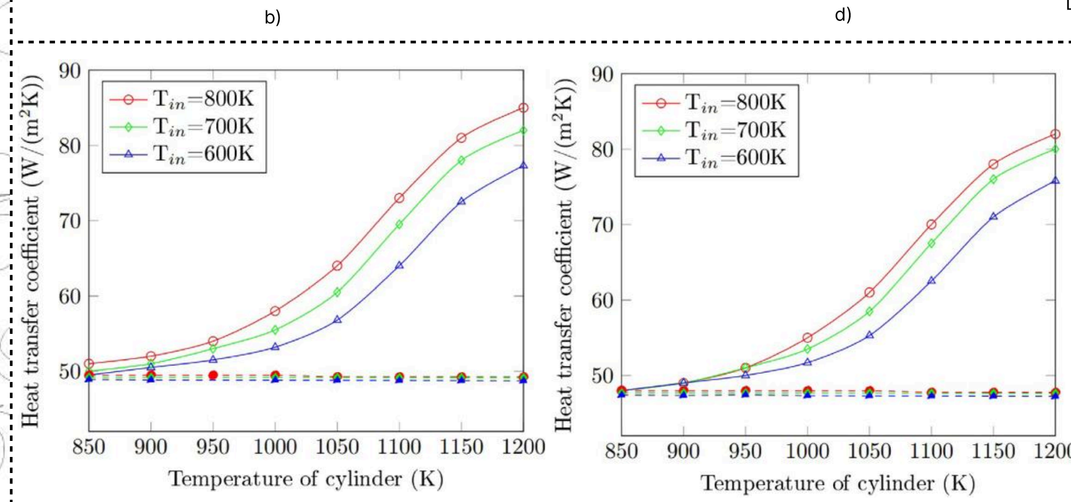




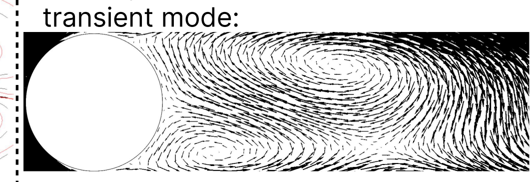
Results



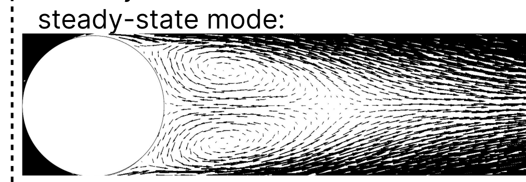




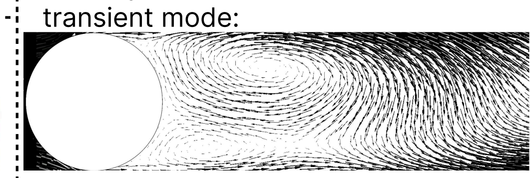
Velocity vectors for the reactive transient mode:



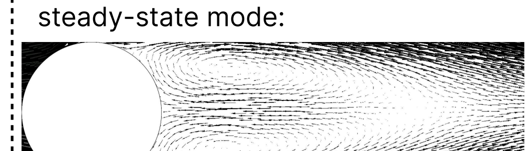
Velocity vectors for the reactive steady-state mode:



Velocity vectors for the non-reactive transient mode:



Velocity vectors for the non-reactive steady-state mode:



Conclusions

At a catalytic cylinder temperature of 1200 K, vortex size and shedding period increase maximally due to high reaction rates and reduced gas density from product formation.

Endothermic SMR reactions significantly elevate heat transfer between the catalyst and flow. The heat transfer coefficient exhibits a nonlinear temperature dependence, peaking at 1200 K due to intense heat absorption.

Vortex formation occurs faster, and local flow velocity is higher in transient (unsteady) regimes compared to steady-state conditions. This amplifies thermal and mass transfer, particularly in reactive flows.